Gas Absorption

Final Lab Report
Unit Operations Lab 2
August 6, 2026

Group 1
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1. Executive Summary (Examples at end of the Guideline) (10 pts)

The objective of this experiment was to evaluate the hydraulic behavior and carbon dioxide absorption performance of a packed-bed gas absorber using deionized water, and to compare open-loop and closed-loop operation for potential gas treatment and carbon capture applications. The packed column was first tested by measuring pressure drop as air flow rate was increased at several water flow rates in order to identify flooding behavior. Absorption performance was then evaluated by measuring dissolved CO2 in liquid samples through NaOH titration and by measuring outlet gas CO2 concentration with a Hempl apparatus.

The pressure drop across the packed bed increased with increasing air and water flow rate, and flooding occurred at lower air flow rates when the water flow rate was higher. In the closed-loop absorption test, dissolved CO2 concentration increased with time and approached a steady value of about 0.001 M after approximately 5 minutes. When the CO2 supply was turned off, the concentration decreased to about 0.00005 M in about 5 minutes. In the open-loop system, the dissolved CO2 concentration increased more gradually and reached a higher steady concentration of about 0.003 M after roughly 17 minutes. The outlet gas CO2 volume fractions were measured as 0.07 for the closed-loop system and 0.05 for the open-loop system. The average mass transfer coefficients were reported as 0.0011 for Experiment 2a, -0.00055 for Experiment 2b, and 0.01635 for the open-loop experiment.

Overall, the results showed that increasing liquid flow increased pressure drop and caused the column to flood sooner. The open-loop system provided the best CO2 absorption performance because fresh water continuously maintained a stronger driving force for mass transfer. Although one negative mass transfer coefficient suggested experimental inconsistency, the results overall support the conclusion that open-loop operation is more effective for industrial gas absorption and carbon capture service.

2. Introduction (10 pts)

Gas absorption is an important separation process used to remove a soluble component from a gas stream by contacting it with a liquid absorbent. (1) In this experiment, a packed-bed absorber was used to evaluate the removal of carbon dioxide from an air stream using deionized water as the absorbent. The objective of the experiment was to study the hydraulic behavior of the packed column and to determine the effectiveness of CO2 transfer from the gas phase to the liquid phase under different operating conditions. (2) This type of operation is relevant to many industrial processes, especially gas treatment and carbon capture systems, where efficient mass transfer and stable column operation are essential. The performance of the absorber depends strongly on gas-liquid contact, interfacial area, residence time, and the mass-transfer driving force. The void fraction of the packing is also an important parameter because it influences the available flow area, fluid holdup, and contact efficiency within the column.

The experimental procedure consisted of two main parts. First, the pressure drop across the packed bed was measured as the air flow rate was increased at different water flow rates to observe column behavior and identify flooding conditions. Second, CO2 absorption performance was evaluated in both closed-loop and open-loop water systems. Dissolved carbon dioxide in liquid samples was determined by titration with sodium hydroxide, while the CO2 concentration in the exiting gas stream was measured using a Hempl apparatus. These measurements were then used to compare the absorption behavior of the two operating modes and to estimate overall mass transfer performance. By relating pressure drop behavior to flooding and concentration data, the experiment provides a practical understanding of how packed absorbers operate and why they are widely used in industrial separation processes.

3. Theory (10 pts)

This experiment investigates how well a packed-bed absorber is at removing carbon dioxide (C) from the air stream using DI water as the absorbent. Gas absorption is commonly used in separation processes, utilizing interface mass transfer to remove a solute, in our case carbon dioxide, from the gas phase into the liquid (water). A very common industry that uses this is the carbon capture industry. The mass transfer of carbon dioxide into water is affected by contact efficiency. This experiment will show the results of CO2 absorption by water in a closed-loop and an open-loop water system.

The Void fraction () is a geometric property used in packed bed systems, which is defined as the ratio of the void column to the total volume. This geometric property is important to note, as it affects the residence time, the flow, and the available interface area for mass transfer.

- is the volume through which the liquid is allowed to flow
- is the total volume of the column

When gas and liquid flow in opposite directions through the packed bed, the flow resistance causes a pressure drop. The pressure drop across the column can determine the flow behavior and indicate when the column is flooding, which occurs when gas flow is relatively high, leading to an increase in pressure drop.

To measure the volume fraction of carbon dioxide in the air leaving the column in the gas stream a Hempl apparatus is used to do so.

To measure the amount of carbon dioxide in water, sodium hydroxide will be used as the CO2 in the water does form a acid. The amount of carbon dioxide can be calculated using the following equation.

Where Cd is the concentration of free dissolved CO2 which can be found by using the following relation

Where Vb is the volume of the titrant (NaOH), [NaOH] is the concentration, and Vw is the volume of the sample. This relation works because NaOH neutralizes carbonic acid in a 1:1 ratio; by finding the amount of NaOH needed to neutralize the carbonic acid, you can determine the amount of carbon dioxide dissolved in the water.

The expression for carbon dioxide transported across the gas-liquid interface is as follows.

Where N is the moles of CO_2 transferred, ka is the mass transfer coefficient multiplied by the interfacial area, and delta c is the driving force; in our case, the driving force is concentration.

4. Results (15 pts)

The first experiment consisted of determining the pressure drop for air flow rates while keeping the water flow rate constant, which was done at different water flow rates (3L/min,6L/min,8L/min)

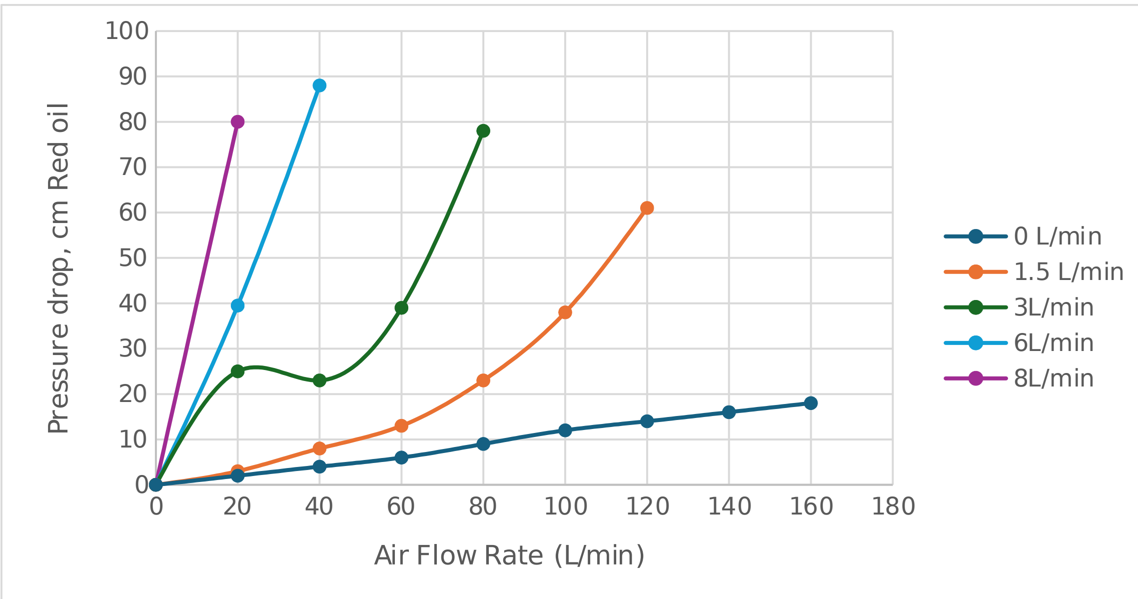

Figure 1: Pressure drop Across the Packed Bed Vs Air flowrates at different Water flow rates

Figure 1 shows the pressure drop across the packed bed as a function of air flow rate for different water flow rates. From Figure 1, the pressure drop increases with increasing water flow rate. This behavior is to be expected: higher liquid flow increases liquid holdup in the column, decreasing the available gas flow area and increasing liquid-gas interaction. As flooding restricts airflow, the gas is forced to flow through narrower passages, increasing frictional resistance and gas-liquid interaction. Consequently, a rapid rise in pressure indicates flooding has occurred, as the column becomes hydraulically overloaded with liquid. It can also be noted that a lower airflow rate is required to achieve flooding at higher water flow rates, as expected. So, in summary, for higher water-flow-rate systems, higher air pressure is required to push through, but less airflow is needed to achieve flooding.

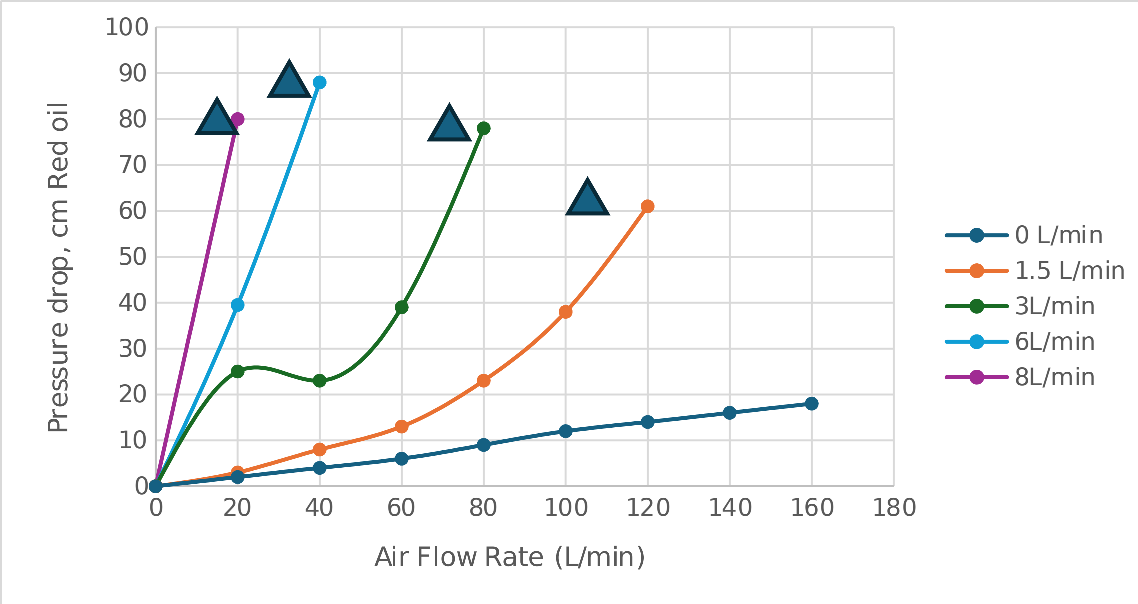

Figure 2: Pressure drop VS Air flow

Figure 2 shows when flooding has been achieved at the different water flows, as indicated by the triangular shape.

Close loop – Absorption

For the Closed-loop absorption, the concentration of CO2 was measured from multiple 50 mL solution samples. The liquid flow rate was kept at 1.5L/min. Allowing CO2 to enter the system in a closed loop allowed for the absorption to be measured. The dissolved Carbon dioxide was then neutralized with NaOH.

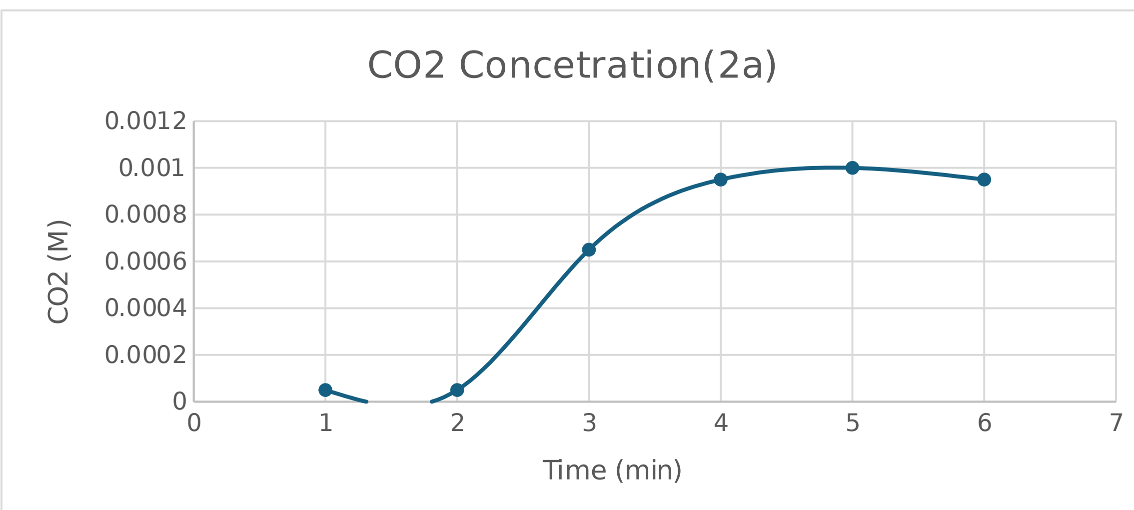

Figure 3: CO2 concentration Vs Time (closed loop part A)

Figure 3 shows the relationship between the CO2 concentration and time. From Figure 3, it can be observed that, over time, the CO2 concentration in the 50 mL samples increased steadily until the 5 min mark, at which point it reached a steady-state value.

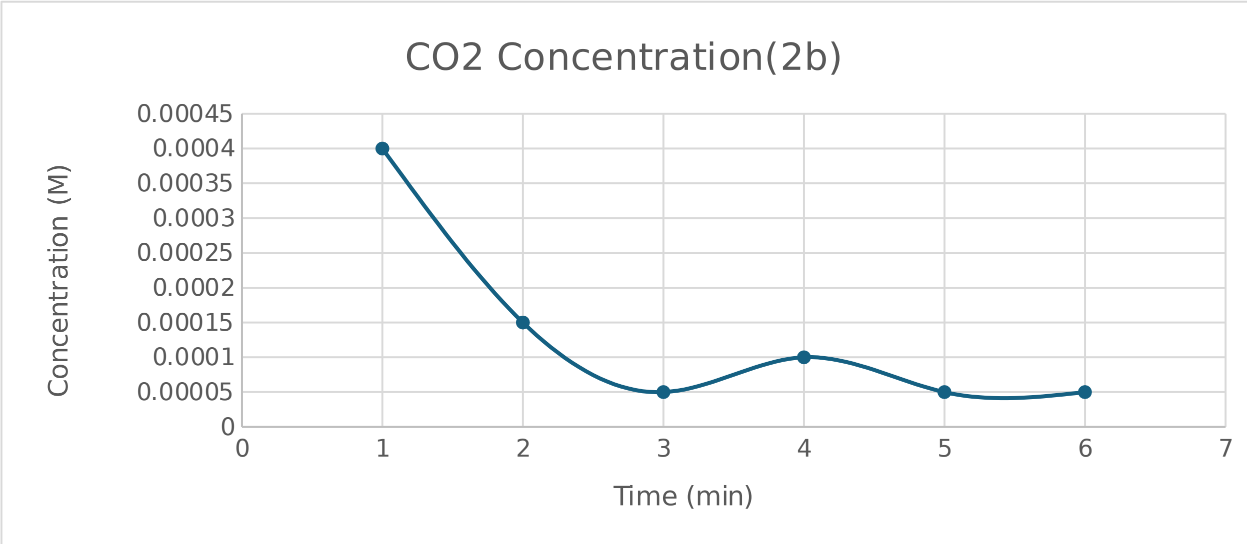

Figure 4: CO2 concentration Vs Time (Closed loop)

Figure 4 shows the relationship between CO2 concentration and time; however, in experiment 2b, the CO2 flow was turned off. Because the CO2 flow was turned off, the CO2 concentration steadily decreased until it reached a steady-state value of 0.00005 M. The desorption took about 5 minutes to reach steady state.

Open Loop Absorption

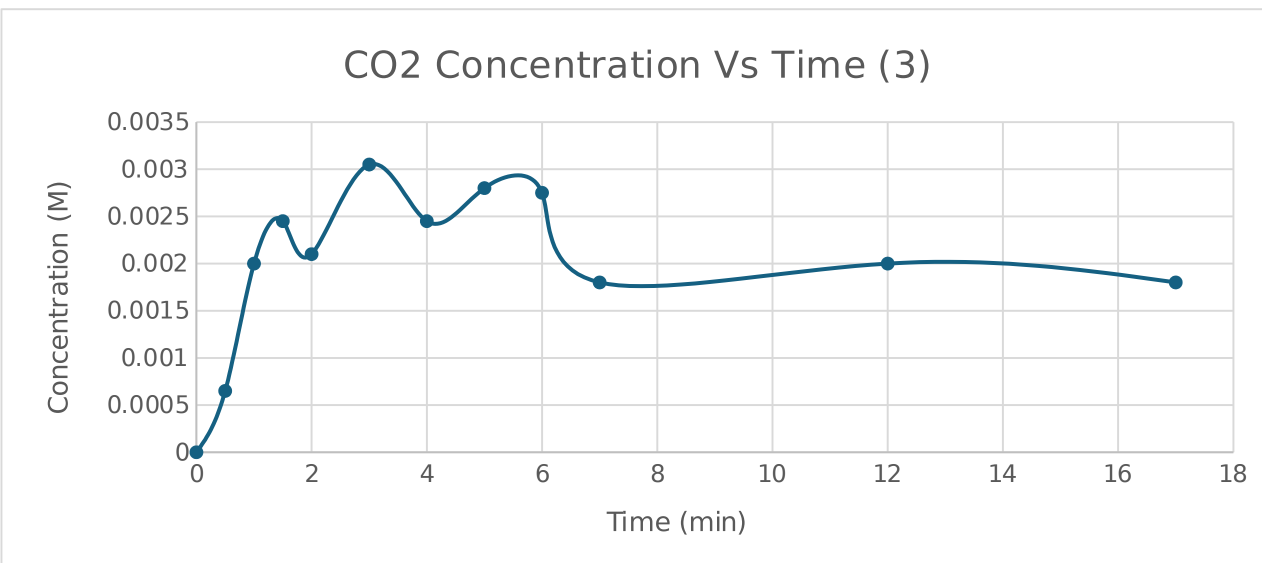

Figure 5: CO2 Concentration Vs Time (open loop)

Figure 5 shows the relationship between CO2 concentration and time in an open-loop system. The CO2 concentration in the samples increased with time until it reached a maximum of 0.003 M; however, it took longer to reach steady state than in the closed-loop experiment. Figure 5 shows a steady-state value at roughly 17 minutes, which is possible because an open system receives constant fresh water, disrupting CO2 concentrations.

In the open-loop system, steady state was reached more slowly because fresh water continuously entered the column while CO2-rich water was drained. This constant dilution prevented the rapid buildup of dissolved CO2, unlike the closed-loop system in part 2, where the same water recirculated and approached equilibrium more quickly.

CO2 Concentration in the exiting gas

A Hempl Apparatus was used to measure CO2 concentration in the existing gas for both the closed- and open-loop experiments.

Table 1: Volume Fraction and Molar Concentration of CO2 in the air

	Volume fraction		Concentration (M)
Outlet closed loop	0.07		0.006757132
Outlet Open loop	0.05		0.004826523

Table 1 shows the volume fraction and Molar concertation of the closed loop and open loop. The outlet closed loop is greater than that of the outlet open loop, and so is the concentration. This is to be expected with how non-recycled water from the open loop allowed for the "cleaner" water to be introduced to the system essentially increasing the overall concentration of CO2 in the water compared to that of the closed loop, hence lower concentration of CO2 in the air.

Overall Mass Transfer Coefficients

Table 2 : Average Mass Transfer coefficients

Experiment	Mass transfer coefficient Ka
2a	0.0011
2b	-0.00055
3	0.01635

Table 2 shows the average mass transfer coefficients of each experiment. The experiment with the highest mass transfer coefficient suggests that an open-loop system has a faster rate of CO2 transfer to DI water and enhanced efficiency. Although an open loop takes longer to reach a steady-state concentration, it still shows that it's a better way to transfer CO2 gas to DI water. This can be explained by the concentration driving force being greater in an open-loop system, where there is a constant feed of fresh water, thereby increasing overall mass transfer.

5. Discussion (15 pts)

The results of the experiment align well with the existing theory of gas absorption in a packed bed. In the lab manual theory, the rate of CO2 absorption is determined by the driving force and the overall mass transfer coefficient described by the relationship, .

The relationship observed in CO2 transport was evident in the comparison between the open-loop and closed-loop systems. As the results show, higher mass transfer coefficients are observed in the open-loop system when fresh water is continuously input, rather than when the same water is recirculated. The addition of fresh water increased the concentration gradient and the driving force for CO2 (3). In closed-loop systems, equilibrium was reached more quickly because recirculation continuously increased CO2 concentration rather than diminishing over time.

The plots in Figures 3 and 4 compare the CO2 concentration in the open-loop V and the closed-loop. The rate of increasing concentration in the closed-loop is almost equal to the rate of diminishing CO2 concentration in the open-loop system. In the closed loop, at around 5 minutes, a plateau is observed with a steady-state value of 0.001 M. This plateau corresponds to the system's equilibrium point. In the open-loop system, the concentration after 3 minutes reached a steady-state value of 0.00005 M, indicating that the diminishing CO2 concentration reached steady state more quickly than the increasing CO2 concentration. This is due to the continuous addition of fresh water, which has caused dilution (3).

As the air flow rate increased, the pressure drop gradually increased, and near the flooding point, the differential pressure reading showed a significant shift. Flooding occurs when high vapor velocity prevents liquid from flowing downward, leading to accumulation, increased pressure drop, and reduced separation efficiency (2). When the flow rate increased, the flooding point was reached much more quickly as the voids filled more rapidly, resulting in higher liquid holdups.

Several assumptions were made in analyzing this experiment. One assumption is ideal mixing in the liquid phase and minimal gas resistance. The assumption was made to simplify the calculations, but it may not allow them to model the experiment correctly, as mass transfer involves both gas and liquid phases to quantify it accurately. Moreover, a negative mass transfer coefficient was calculated for experiment 2b; this is quite a meaningless number. This can be caused by an inconsistency in the product sampling or by possible fluctuations in air and water flow.

The conclusions from this experiment are that as gas and liquid flow rates increased in the system, the pressure drop increased. Increasing the liquid flow rate shifted the flooding point to a higher value, as a stronger driving force was introduced. Open-loop systems have a higher mass transfer due to a sustained driving force. The higher the open-loop system's efficiency, the more effective it is for industrial-scale carbon capture.

A few questions that were left unattended, such as the relative contributions of the gas and liquid phases to mass transfer. The experiment can vary depending on the type of packing and material used to pack the bed, as well as the temperature conditions in the bed.

6. References (10 pts)

(1) Armfield Ltd. *Gas Absorption Column–UOP7*; Issue 11; Armfield Ltd.: Ringwood, Hampshire, England, 2009.Green, D. W.; Southard, M. Z., Eds.

(2) *Perry's Chemical Engineers' Handbook*, 9th ed.; McGraw-Hill Education: New York, 2018.
(3) Mombourquette, M. Appendix 1: Propagation of Errors in Calculations. **2020**.
(4) *Flooding Discussion – Online Unit Operations Laboratory*. https://unitops.chem-eng.utoronto.ca/flooding-discussion/ (accessed 2026-04-05).
(5) *The Ultimate Guide to Gas Absorption*. https://www.gustawater.com/blog/gas-absorption.html (accessed 2026-04-05).

7. Appendix I: Data Tabulation/Graphs (10 pts)

Table 3: Void Fraction Calculation

beaker, g	beaker with 60 ml rr, g	with water, g	rr, g	V void mL	V total mL	Void Fraction
103.7	127	176	54.7	42	53	7.92E-01

Table 4: Experiment B Data

Water Flow LPM	Air Flow LPM	dP cmOil
	0	0
0	20	2
	40	4
	60	6
	80	9
	100	12
	120	14
	140	16
	160	18
	180	-

Water Flow LPM	Air Flow LPM	cmOil
	0	0
1.5	20	3
	40	8
	60	13
	80	23
	100	38
	120	61
	140	-
	160	-
	180	-

Water Flow LPM	Air Flow LPM	dP cmOil
1.5	20	
	40	
	60	
	80	

	100	
	120	
	140	
	160	
	180	

Water Flow LPM	Air Flow LPM	dP cmOil
	0	0
3	20	25
	40	23
	60	39
	80	78
	100	-
	120	-
	140	-
	160	-
	180	-

Water Flow LPM	Air Flow LPM	dP cmOil
	0	0
6	20	39.5
	40	88
	60	-
	80	-
	100	-
	120	-
	140	-
	160	-
	180	-

Water Flow LPM	Air Flow LPM	dP cmOil
	0	0
8	20	80
	40	-
	60	-
	80	-
	100	-
	120	-
	140	-
	160	-
	180	-

Table 5: Titration Data

sample #	mL start	mL end	NaOH added(mL)	NaOH added(L)	Moles of NaOH
0 (init)	0	0.1	0.1	1.00E-04	5.00E-06
1	0.1	0.2	0.1	1.00E-04	5.00E-06
2	0.2	1.5	1.3	1.30E-03	6.50E-05
3	1.5	3.4	1.9	1.90E-03	9.50E-05
4	3.4	5.4	2	2.00E-03	1.00E-04
5	5.4	7.3	1.9	1.90E-03	9.50E-05
6	7.3			0	0
7	0			0	0

Table 6: CO2 Data

Moles of Co2	Co2 Concentration inlet (in smaple)	Delta C
2.50E-06	5.00E-05	6.71E-03
2.50E-06	5.00E-05	6.71E-03
3.25E-05	6.50E-04	6.11E-03
4.75E-05	9.50E-04	5.81E-03
5.00E-05	1.00E-03	5.76E-03
4.75E-05	9.50E-04	5.81E-03
0	0	
0	0	

Table 7: Calculations of N and K

N	K		
0.00E+00	0.00E+00	Volume Air Sample (mL)	2.80E+00
0.00E+00	0.00E+00	Volume of cylinder (cc)	4.00E+01
1.00E-05	1.64E-03	Volume Fraction Co2	7.00E-02
1.13E-05	1.94E-03	Ptotal PSI	3.47E+01
9.50E-06	1.65E-03	Ptotal atm	2.36E+00
7.50E-06	1.29E-03	Pco in	1.65E-01
		Concentration of Co2 outlet	6.76E-03

Table 8: Titration Data

sample #	mL start	mL end	NaOH added (mL)	NaoH added (L)	Moles of NaoOH
0 (init)	0	3.8			

1	3.8	4.6	0.8	8.00E-04	4.00E-05
2	4.6	4.9	0.3	3.00E-04	1.50E-05
3	4.9	5	0.1	1.00E-04	5.00E-06
4	5	5.2	0.2	2.00E-04	1.00E-05
5	5.2	5.3	0.1	1.00E-04	5.00E-06
6	5.3	5.4	0.1	1.00E-04	5.00E-06

Table 9: CO2 Data

Moles Of CO2'	Co2 Concentration inlet (in smaple)	Delta C
2.00E-05	4.00E-04	6.36E-03
7.50E-06	1.50E-04	6.61E-03
2.50E-06	5.00E-05	6.71E-03
5.00E-06	1.00E-04	6.66E-03
2.50E-06	5.00E-05	6.71E-03
2.50E-06	5.00E-05	6.71E-03

Table 10: N and K Calculations

N	K
0.00E+00	0.00E+00
-6.25E-06	-9.46E-04
-5.83E-06	-8.70E-04
-3.75E-06	-5.63E-04
-3.50E-06	-5.22E-04
-2.92E-06	-4.35E-04

Table 11: Trial 3 Data

sample #	mL start	mL end	NaOH added(mL)	NaoH added L	
0 (init)	0	5.4	0	0.00E+00	30s
1	5.4	6.7	1.3	1.30E-03	
2	6.7	10.7	4	4.00E-03	
3	10.7	15.6	4.9	4.90E-03	
4	15.6	19.8	4.2	4.20E-03	
5	19.8	25.9	6.1	6.10E-03	1min
6	25.9	30.8	4.9	4.90E-03	
7	30.8	36.4	5.6	5.60E-03	
8	36.4	41.9	5.5	5.50E-03	
9	41.9	45.5	3.6	3.60E-03	
10	45.5	49.5	4	4.00E-03	5 min
		45.9	3.6	3.60E-03	

Table 12: NaOH and CO2 Data

Moles of NaOH	Moles of Co2	Delta C
0.00E+00	0.00E+00	4.83E-03
6.50E-05	3.25E-05	4.18E-03
2.00E-04	1.00E-04	2.83E-03
2.45E-04	1.23E-04	2.38E-03
2.10E-04	1.05E-04	2.73E-03
3.05E-04	1.53E-04	1.78E-03
2.45E-04	1.23E-04	2.38E-03
2.80E-04	1.40E-04	2.03E-03
2.75E-04	1.38E-04	2.08E-03
1.80E-04	9.00E-05	3.03E-03
2.00E-04	1.00E-04	2.83E-03
1.80E-04	9.00E-05	3.03E-03

Table 13: N, K, and CO2 calculations

N	K	Volume Air Sample (mL)	2
#DIV/0!	#DIV/0!	Volume of cylinder (cc)	40
6.50E-05	1.56E-02	Volume Fraction	0.05
1.00E-04	3.54E-02	P total PSI	34.7
8.17E-05	3.44E-02	P total atm	2.36E+00
5.25E-05	1.93E-02	P CO2 in	1.18E-01
5.08E-05	2.86E-02		4.83E-03
3.06E-05	1.29E-02		
2.80E-05	1.38E-02		
2.29E-05	1.10E-02		
1.29E-05	4.25E-03		
8.33E-06	2.95E-03		
5.29E-06	1.75E-03		

8. Appendix II: Error Analysis (10 pts)

<u>Quantitative</u>

A 50 mL Burette has an uncertainty of 0.05 mL at 25 C (1),

5.6 mL is the highest amount of titrant added in experiment 3,

All burettes, depending on the volume, have a degree of uncertainty that can cause inaccuracies in the concentration, depending on the amount of titrant added. The smaller the amount added, the greater the error due to uncertainty.

0.1 mL was the smallest amount of titrant recorded in experiment 2A. This shows how a small amount can lead to a potentially large error in the determination of CO2 concentration.

Qualitative

Errors can occur when a column leaks from the seals on the sides, causing fluctuations in the packed column. Thus, it can affect the CO2 concentration gradient, as the sample was leaking out of the side of the column at an unknown rate. Moreover, the driving force was reduced because the system was not perfectly sealed, resulting in pressure loss and uneven wetting of the column. This can inherently affect the flooding point reached, and the system's reaching steady state could have been delayed, causing deviations in the time to steady state.

9. Appendix III: Sample Calculations (10 pts)

Part 1

The first step was to calculate the void volume of the packed bed using the following equation:

Where:

Using our data:

The next part was to calculate the pressure drop. The calculation shown below uses data with a water flow rate of 1.5 LPM and an air flowrate of 40 LPM. This was done by using the following equation:

Where:

Plugging values in:

This was done for all trials.

Part 2a

First, the CO2 concentration will be calculated from the sample line.

Data is taken from the third sample. To find the volume of added titrant, the following is used:

Convert to moles using 0.05 N:

From here, the balanced chemical reaction equation is used to find moles of CO2.

The chemical reaction is:

CO_2 concentration for 5 minutes:

This was done for all trials in 2a, alongside all trials in 2b and 3.

Next, the CO_2 concentration is found using the Hempl Apparatus.

The volume of the air sample is 2.8 mL and the volume of the cylinder is 40 cc.

This procedure was repeated for experiment 3 as well.

Lastly, the mass transfer coefficient is found for 5 minutes. The expression is:

Solving for ka:

N can be found be using the following:

Next, is found:

ka is now found:

This is repeated for all trials.

10. Appendix IV: Individual Team Contributions

NAME: Adil Feratovic

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Attended lab
Pre-Lab Editing	1	Edited procedure
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	3	Worked on this part
Project Safety Evaluation (prelab)	0	
Final Lab Editing	2	Edited parts worked on
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	1	Added used references
Data Tabulation/Graphs (final)	3	Worked on data tabulation
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	3	Did calculations
Power Point Presentation	1	
Total	19	

NAME: Aaron Casas

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Attended lab sessions
Pre-Lab Editing		
Experimental Objective (Prelab)	0	

Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	.5	Completed safety portion of the safety prelab.
Final Lab Editing	2	Edited theory and results section.
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	2	Explained how and what equations where used to calculated the numerical values.
Results (final lab)	4	Included data and graph figures explaining the trends in data.
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	3	Calculated values, tabulated data, and formed graphs
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	.5	Worked on safety and Challenges slide
Total	15	

NAME: Anna Cai

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	4	Attended the CCP meeting
Pre-Lab Editing	1	
Experimental Objective (Prelab)	.5	
Apparatus (prelab)	1	Wrote descriptions
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	1	Fixed grammar

Executive Summary (final lab)	1	
Introduction (Final lab)	1	
Theory (Final Lab)	1	Edited
Results (final lab)	3	Completed calculations
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	.5	
Data Tabulation/Graphs (final)	1	Formatted tables
Error Analysis (final lab)		
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Apparatus and materials
Total	16	

NAME: Hussein Rezk

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	4	Operated the stirred tank
Pre-Lab Editing	1	Experimental
Experimental Objective (Prelab)	1	Experimental
Apparatus (prelab)		
Materials & Supplies (prelab)	1	Listed required material
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	5	Discussed results and correlations
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)	2	Conducted qualitative and quantitative analysis
Sample Calculations (final		

Lab)		
Power Point Presentation	2	Theory sections
Total	16	